Psychological Assessment
2015, Vol. 27, No. 4, 1500–1506


1040-3590/15/$12.00 http://dx.doi.org/10.1037/pas0000126

BRIEF REPORT

# Suicide Risk Screening: Comparing the Beck Depression Inventory-II, Beck Hopelessness Scale, and Psychache Scale in Undergraduates

Talia Troister, Madeleine T. D'Agata, and Ronald R. Holden
Queen's University

The current research evaluates the effectiveness and relative merits of 3 screening measures (the Beck Depression Inventory-II, the Beck Hopelessness Scale, and the Psychache Scale) in evaluating preexisting suicide risk factors for a sample of 7,522 undergraduate students. All measures demonstrated significant diagnostic accuracy for indicating suicide ideation, previous single and multiple suicide attempts, and a recent suicide attempt, which are all serious risk factors for subsequent death by suicide in university students. However, the Psychache Scale displayed superior performance in accurately identifying suicide risk compared with both the Beck Depression Inventory-II and the Beck Hopelessness Scale. Identifying students most at risk for suicide requires diagnostically efficient measures, thus preliminary cut-scores for identifying at-risk students are provided.

*Keywords:* suicidal behavior, screening measures, depression, hopelessness, psychache



With death by suicide being a societal issue, understanding and predicting suicidal behavior is an important public health goal. Globally, nearly 1 million people die by suicide every year (National Institute of Mental Health, 2009). Suicide is one of the 12 primary causes of death in the United States and it claims the lives of more than 100 individuals in the U.S. each day (National Center for Health Statistics, 2014).

For universities, suicide is a particular concern because it is the third primary cause of death in the U.S. college age group. The American College Health Association (2000) indicated that 9.5% of students had seriously considered suicide in the previous year and 1.5% had attempted to die by suicide. In a study of over 26,000 students at 70 U.S. colleges, 6% of undergraduate and 4% of graduate students had seriously considered suicide and 0.85% of undergraduate and 0.30% of graduate students had attempted to die by suicide in the previous 12 months (Drum, Brownson, Burton Denmark, & Smith, 2009).

A primary risk factor for suicide is previous suicidal behavior (Moscicki, 1997). Increased risk is linked to suicide ideation, a previous attempt (Rosenberg et al., 1988; Roy, Segal, Centerwall, & Robinette, 1991), recent suicidal communications (Robins, Murphy, Wilkinson, Gassner, & Kayes, 1959), and previous multiple attempts (Rudd, Joiner, & Rajad, 1996). The robustness of past suicidality for current risk remains even when covarying out psychiatric syndrome and suicide-related personality factors (e.g., hopelessness; Joiner et al., 2005). It should be recognized, however, that suicide ideation, previous, recent previous, and multiple previous suicide attempts, although health issues themselves and relevant risk factors for suicide, are nevertheless, fallible proxies for death by suicide. Further, distinguishing between risk factors and indices predictive of suicide is essential because assessing risk does not necessarily result in effective prediction or intervention (Paris, 2006; Roos, Sareen, & Bolton, 2013).

Although behavioral and demographic variables (e.g., sex) can indicate suicide risk, identifying psychological risk factors is important because such variables may be therapeutically malleable (Brown, Beck, Steer, & Grisham, 2000). Assessing psychological factors may also be more publicly acceptable with nonclinical populations (e.g., students) than asking directly about serious suicidal behavior, can help overcome inhibitions that some health care professionals have in inquiring directly about suicide, and may address issues of concealment of plans and urges for suicide in some individuals such as perfectionists (Flett, Hewitt, & Heisel, 2014; Friedman, 2006). Psychological factors with consistent relevance for suicidal behavior include depression (Thomson, 2012), hopelessness (Brown et al., 2000), and psychache (i.e., psychological pain/anguish/despair; Troister & Holden, 2012). Depression, hopelessness, and psychache overlap but are distinct constructs (DeLisle & Holden, 2009; Troister & Holden, 2013).

Operational screens for depression, hopelessness, and psychache include the Beck Depression Inventory—II (BDI-II; Beck, Steer, & Brown, 1996), the Beck Hopelessness Scale (BHS; Beck, Weismann, Lester, & Trexler, 1974), and the Psychache Scale (PAS; Holden, Mehta, Cunningham, & McLeod, 2001). These instruments are short, self-report measures whose test scores and interpretations have shown

This article was published Online First April 27, 2015.

Talia Troister, Madeleine T. D'Agata, and Ronald R. Holden, Department of Psychology, Queen's University.

This research was supported in part by grants from the Social Sciences and Humanities Research Council of Canada.

Ronald R. Holden is the author copyright holder of the Psychache Scale.

Correspondence concerning this article should be addressed to Ronald R. Holden, Department of Psychology, 62 Arch Street, Queen's University, Kingston, Ontario K7L 3N6, Canada. E-mail: holdenr@queensu.ca

empirical reliability and validity for various populations in statistically predicting suicidal manifestations (Beck, Steer, & Carbin, 1988; Durham, 1982; Mills, Green, & Reddon, 2005). For example, in a prospective study of 1958 psychiatric outpatients for an average of 43 months, Beck, Brown, Berchick, Stewart, and Steer (1990) reported large effect sizes for the BDI and BHS in predicting ultimate suicide. In a 2-year prospective study of 41 suicide ideators and attempters, Troister and Holden (2012) found not only did the PAS predict suicide ideation and change in ideation, it had added value relative to using only the BDI and BHS.

Although research has indicated that measures of depression, hopelessness, and psychache can each have added value for statistically predicting suicide criteria (Troister & Holden, 2013), a direct comparison among operationalizations has not been undertaken. Thus, with the goal of identifying the single optimal brief measure for identifying suicide risk among these three factors, we comparatively evaluate the BDI-II, BHS, and PAS for their respective abilities to individually screen for suicide at-risk university students.

# Method

## Participants and Procedure

This study was approved by a University Research Ethics Board. Between 2008 and 2012, 7,522 undergraduates (5,381 women, 1,784 men, 357 unreported) at a midsize Canadian university (>80% Caucasian) were recruited from an introductory psychology subject pool during the first month of the academic year. Participants had a mean age of 18.23 years ($SD$ = 1.93, Median = 18.00). Individuals completed the materials listed below in a package that contained a consent form and a debriefing sheet with information on local counseling resources.

## Materials

**Screening scales.** The BDI-II (Beck et al., 1996) is a 21-item, self-report scale of depression severity. Items are answered on 4-point ratings. Scale scores have high internal reliability for psychiatric outpatients (Beck, Steer, Ball, & Ranieri, 1996) and convergent and discriminant validity in a sample of psychiatric inpatients (Beck et al., 1996). In a 20-year study of psychiatric outpatients, Brown, Beck, Steer, and Grisham (2000) showed the validity of the BDI scores for predicting eventual suicide. The BHS (Beck et al., 1974) is a 20-item true/false self-report scale that measures hopelessness about the future. The authors have reported a test score alpha reliability for suicide attempters of .93. Validity for BHS scores has been shown with their ability to predict suicide among almost 2,000 consecutive psychiatric outpatients 5 to 12 years later (Beck, Brown, Berchick, Stewart, & Steer, 1990). The PAS (Holden et al., 2001) comprises 13 self-report items answered on 5-point ratings (see Appendix) that index Shneidman's (1993) concept of psychache/psychological pain. Coefficient alpha reliabilities over .90 for PAS scores are reported for university (Troister & Holden, 2010) and offender (Mills et al., 2005) samples. Validity has been shown through the PAS scores' ability to distinguish suicide attempters from nonattempters (Holden et al., 2001).

**At-risk groups.** Participants indicated if they had ever attempted suicide, and if so, when the most recent attempt was, and the number of lifetime attempts. Individuals also answered the Beck Scale for Suicide Ideation (BSS; Beck & Steer, 1993). Beck and Steer (p. 7) designate respondents as suicide ideators if they provide nonzero BSS answers for Item 4 (active intention) or Item 5 (nonavoidance of death). Based on these reports, participants were classified into four nonmutually exclusive at-risk groupings: suicide ideator versus nonideator; lifetime single suicide attempter versus nonattempter; lifetime multiple suicide attempter versus lifetime nonmultiple attempter; recent suicide attempter (past year) versus not a recent suicide attempter. Further, an elevated-risk group was designated—suicide ideators with lifetime multiple suicide attempts including an attempt in the past year versus others. Groups were not mutually exclusive because these groupings have been individually identified in the literature as being a relevant risk factor (Robins et al., 1959; Rosenberg et al., 1988; Roy et al., 1991; Rudd et al., 1996) and any overlap has ecological validity in that these are naturally occurring overlapping clusters.

Proportions of women and men did not differ among groups with regard to suicide ideation, a recent suicide attempt, or being in the elevated-risk group ($ps > .05$), however, marginal effects were present for lifetime suicide attempts ($p = .03$), with those reporting attempts differentially more likely to be women. Groupings were not related to age ($ps > .05$) but this was not unexpected because all participants were first-year university course students. Given these minor associations for demographics, they were not considered further in the analyses.

## Data Analysis

Merits of the BDI-II, BHS, and PAS for indicating risk were primarily evaluated with receiver operating characteristic (ROC) curves. These analyses consider sensitivity (i.e., true positive rate) and 1–specificity (i.e., false alarm rate) and are widely accepted for obtaining optimal test cut-scores and for comparing diagnostic accuracy among tests (Akobeng, 2007b). ROC analyses do not assume data normality or homoscedasticity and are independent of base rates and established cut-off scores. For our non-normally distributed variables, this method of analysis was preferred. ROC informational value is summarized as an AUC (area under curve) that is compared with a chance value of .50. The AUC is the probability that a pair of cases selected randomly from two underlying distributions will be classified correctly. AUCs of .50–.70, .70–.90, or over .90 are low, moderate, or high accuracy, respectively (Streiner & Cairney, 2007). AUCs and differences in AUCs can be tested for statistical significance (Hanley & McNeil, 1983). In comparing two AUCs for the same cases, Hanley and McNeil (1983) indicate that a $z$ statistic represents the difference between areas relative to a pooled error comprising individual AUC standard errors as well as the correlation between areas.

Positive prediction power (PPP), negative prediction power (NPP), and relative risk ratios (RRR) were also considered. Whereas PPP is the proportion of cases with a positive test result who actually manifest the diagnosis, NPP is the proportion of cases with a negative test result who do not actually have the diagnosis. RRR is the ratio of the probability of a diagnostic event for a group above a cut-score relative to the probability of the event for a group below that score. In particular, the RRR is a readily interpretable summary indicating how many times a score above a cutpoint is more likely to manifest the target state or condition.

Table 1
*Scale Descriptive Statistics, Coefficient Alpha Reliabilities, and Correlations (N = 7,522)*

| Scale | *M* | *SD* | Skewness | Kurtosis | Scale correlations 1 | 2 | 3 |
|---|---|---|---|---|---|---|---|
| 1. Beck Depression Inventory-II | 8.03 | 7.16 | 1.64 | 3.75 | .90 | | |
| 2. Beck Hopelessness Scale | 3.22 | 3.04 | 1.95 | 5.01 | .52 | .80 | |
| 3. Psychache Scale | 20.35 | 8.25 | 1.76 | 7.47 | .65 | .49 | .94 |

*Note.* Coefficient alpha reliabilities are in the diagonal of the correlations. Values for skewness and kurtosis indicate non-normality.

## Results

Data were screened for accuracy and missing values. For any item, no more than 5% of cases were missing. For participants missing 10% or less of data within any one scale, scale scores were prorated. Participants missing more than 10% of data within a scale had that scale score excluded from analyses. For any particular scale, no more than 5% of cases were excluded.

Descriptive statistics including coefficient alpha reliabilities are in Table 1. Statistics for the BDI-II, BHS, and PAS are similar to previous research with students (DeLisle & Holden, 2009). Scale scores also show strong reliability for our sample. In the current sample, 701 cases (10%) were suicide ideators based on the Beck and Steer (1993, p. 7) criterion, 120 participants (1.6%) had a lifetime single suicide attempt, 96 (1.3%) had lifetime multiple suicide attempts, and 57 (0.83%) indicated a suicide attempt within the past year. Thirty-two cases (0.47%) were designated as elevated risk because they were current suicide ideators, and had a lifetime history of multiple suicide attempts, and had attempted suicide within the previous year. Descriptive statistics for cases in each risk grouping are also shown in Table 2. Relative to the general group, mean scores for the single factor at-risk groups are at least half a standard deviation higher (i.e., a medium effect size) for the BDI-II and BHS, and one standard deviation higher (i.e., more than a large effect size) for the PAS. For the elevated risk group, each scale mean is more than two standard deviations above the general group. Overall, each scale shows psychometric strengths that indicate the potential merit for being an accurate index of suicide risk.

In identifying suicide ideators, AUCs indicated moderate accuracy values of .78, .74, and .79 for the BDI-II, BHS, and PAS, respectively. The BHS AUC was less than for the BDI-II ($z = -3.68$, $p < .01$) and PAS ($z = -4.38$, $p < .01$), the latter two not significantly differing ($z = 1.05$, $p > .05$). In distinguishing lifetime single suicide attempters from nonattempters, AUCs indicated low accuracy for the BDI-II (.67) and BHS (.66), and moderate accuracy for the PAS (.76). The PAS AUC was greater than for the BDI-II ($z = 4.41$, $p < .01$) and BHS ($z = 4.00$, $p < .01$) which did not differ reliably ($z = 0.50$, $p > .05$). In differentiating lifetime multiple suicide attempters from nonmultiple attempters, AUCs indicated low accuracy for the BHS (.65) and moderate accuracy for the BDI-II (.76) and PAS (.80). The BHS AUC was significantly less than for the BDI-II ($z = -3.66$, $p < .01$) and PAS ($z = -4.51$, $p < .01$) and the PAS AUC trended toward being greater than that for the BDI-II ($z = 1.90$, $p = .07$). For classifying recent suicide attempters, AUCs were of low accuracy for the BHS (.69) and of moderate accuracy for the BDI-II (.81) and PAS (.84). The AUC for the BHS was significantly less than for the BDI-II ($z = -3.51$, $p < .01$) and PAS ($z = -4.51$, $p < .01$). The difference between the PAS and BDI-II AUCs trended toward being discrepant ($z = 1.37$, $p = .09$). Finally, in identifying elevated risk individuals (i.e., current suicide ideators with a history of multiple suicide attempts and a suicide attempt in the previous 12 months; see Table 2), AUCs indicated moderate accuracy for the BDI-II (.88) and BHS (.79) and high accuracy for the PAS (.94). All AUCs differed significantly from each other (all $ps < .04$).

In identifying potential BDI-II, BHS, and PAS cut-scores for individual risk groupings, a range of values was considered (see Table 3). For each risk grouping, the optimal cut-score was associated with the PAS. PAS cut-scores of 24, 22, 21, and 25 optimized the combination of sensitivity and specificity for the identification of suicide ideators, lifetime single suicide attempters,

Table 2
*Scale Descriptive Statistics for At-Risk Groups*

| | At-risk group: Suicide ideator ($n = 701$) | | Lifetime single suicide attempt ($n = 120$) | | Lifetime multiple suicide attempts ($n = 96$) | | Suicide attempt in previous 12 months ($n = 57$) | | Elevated-risk group: All risk factors present ($n = 32$) | |
|---|---|---|---|---|---|---|---|---|---|---|
| Scale | *M* | *SD* | *M* | *SD* | *M* | *SD* | *M* | *SD* | *M* | *SD* |
| 1. Beck Depression Inventory-II | 15.72 | 9.92 | 12.85 | 9.27 | 18.77 | 13.47 | 20.64 | 12.87 | 26.00 | 13.70 |
| 2. Beck Hopelessness Scale | 6.29 | 4.69 | 4.83 | 3.81 | 6.47 | 5.77 | 7.36 | 6.22 | 10.08 | 6.79 |
| 3. Psychache Scale | 29.85 | 11.57 | 28.53 | 10.89 | 34.04 | 13.96 | 35.78 | 12.69 | 42.42 | 10.71 |

*Note.* Group membership can overlap.

Table 3
*Diagnostic Efficiency Statistics for Multiple Cut-Scores of Each Screening Scale for Identifying At-Risk Cases*

| | At-risk factor | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Suicide ideator | | | | | Lifetime single suicide attempt | | | | | Lifetime multiple suicide attempts | | | | | Suicide attempt in previous year | | | | |
| Cutpoint | Sensitivity | Specificity | PPP | NPP | RRR | Sensitivity | Specificity | PPP | NPP | RRR | Sensitivity | Specificity | PPP | NPP | RRR | Sensitivity | Specificity | PPP | NPP | RRR |
| BDI-II | | | | | | | | | | | | | | | | | | | | |
| ≥7 | .84 | .55 | .18 | .97 | 5.45 | .71 | .52 | .02 | .99 | 2.60 | .81 | .52 | .02 | 1.00 | 4.52 | .88 | .52 | .01 | 1.00 | 7.67 |
| ≥8 | .79 | .62 | .19 | .96 | 5.30 | .65 | .59 | .01 | .99 | 2.55 | .79 | .58 | .02 | 1.00 | 5.12 | .84 | .59 | .02 | 1.00 | 7.47 |
| ≥9 | .76 | .68 | .21 | .96 | 5.47 | .61 | .64 | .01 | .99 | 2.79 | .77 | .64 | .03 | 1.00 | 5.71 | .83 | .64 | .02 | 1.00 | 8.34 |
| ≥10 | .71 | .74 | .24 | .96 | 5.44 | .56 | .70 | .01 | .99 | 2.84 | .70 | .69 | .03 | .99 | 5.13 | .81 | .70 | .02 | 1.00 | 9.40 |
| ≥11 | .67 | .78 | .25 | .95 | 5.32 | .52 | .74 | .01 | .99 | 3.02 | .67 | .73 | .03 | .99 | 5.40 | .78 | .74 | .02 | 1.00 | 9.34 |
| BHS | | | | | | | | | | | | | | | | | | | | |
| ≥3 | .76 | .55 | .16 | .95 | 3.45 | .70 | .52 | .02 | .99 | 2.47 | .69 | .51 | .02 | .99 | 2.33 | .70 | .52 | .01 | .99 | 2.51 |
| ≥4 | .66 | .72 | .21 | .95 | 4.07 | .55 | .68 | .03 | .99 | 2.65 | .56 | .68 | .02 | .99 | 2.69 | .63 | .68 | .02 | 1.00 | 3.49 |
| ≥5 | .56 | .81 | .26 | .94 | 4.35 | .46 | .78 | .03 | .99 | 3.04 | .54 | .78 | .03 | .99 | 4.09 | .55 | .78 | .02 | .99 | 4.17 |
| ≥6 | .47 | .88 | .31 | .93 | 4.76 | .32 | .85 | .04 | .99 | 2.63 | .44 | .85 | .04 | .99 | 4.23 | .50 | .85 | .03 | .99 | 5.44 |
| ≥7 | .39 | .92 | .35 | .93 | 4.94 | .24 | .89 | .04 | .99 | 2.52 | .35 | .89 | .04 | .99 | 4.16 | .42 | .89 | .03 | .99 | 5.54 |
| PAS | | | | | | | | | | | | | | | | | | | | |
| ≥21 | .75 | .69 | .21 | .96 | 5.36 | .73 | .66 | .03 | .99 | 5.02 | **.80** | **.65** | **.03** | **1.00** | 7.37 | .86 | .65 | .02 | 1.00 | 11.12 |
| ≥22 | .72 | .73 | .23 | .96 | 5.54 | **.71** | **.69** | **.04** | **.99** | 5.39 | .73 | .69 | .03 | .99 | 5.72 | .85 | .68 | .02 | 1.00 | 11.48 |
| ≥23 | .69 | .76 | .25 | .96 | 5.60 | .64 | .73 | .04 | .99 | 4.64 | .72 | .72 | .03 | .99 | 6.37 | .83 | .72 | .03 | 1.00 | 12.01 |
| ≥24 | **.66** | **.79** | **.27** | **.95** | 5.58 | .60 | .76 | .04 | .99 | 4.57 | .68 | .75 | .04 | .99 | 6.15 | .81 | .75 | .03 | 1.00 | 12.71 |
| ≥25 | .62 | .82 | .29 | .95 | 5.65 | .56 | .79 | .04 | .99 | 4.49 | .67 | .78 | .04 | .99 | 6.83 | **.80** | **.78** | **.03** | **1.00** | 13.40 |

*Note.* Within an at-risk grouping, the maximum combination of sensitivity, specificity, positive predictive power, and negative predictive power summed is in bold. PPP = positive predictive power; NPP = negative predictive power; RRR = relative risk ratio.

Table 4
*Diagnostic Efficiency Statistics for Multiple Cut-Scores of Each Screening Scale for Identifying Elevated Risk Cases*

| Cut-score | Sensitivity | Specificity | Positive predictive power | Negative predictive power | Relative risk ratio |
|---|---|---|---|---|---|
| BDI-II ≥ 12 | .82 | .77 | .01 | 1.00 | 15.24 |
| ≥13 | .82 | .80 | .02 | 1.00 | 18.22 |
| ≥14 | .75 | .83 | .02 | 1.00 | 14.13 |
| BHS ≥ 6 | .69 | .85 | .02 | 1.00 | 11.98 |
| ≥7 | .56 | .89 | .02 | 1.00 | 9.77 |
| ≥8 | .56 | .92 | .03 | 1.00 | 13.76 |
| PAS ≥ 26 | .91 | .80 | .02 | 1.00 | 18.18 |
| ≥27 | **.91** | **.82** | **.02** | **1.00** | 44.05 |
| ≥28 | .88 | .84 | .02 | 1.00 | 36.06 |

*Note.* The maximum combination of sensitivity, specificity, positive predictive power, and negative predictive power summed is in bold.

lifetime multiple suicide attempters, and recent suicide attempters, respectively. For identifying elevated suicide risk (see Table 4), a PAS score of 27 or greater yielded the maximum combination of sensitivity and specificity. Associated with this PAS cut-score for identifying elevated risk were positive and negative prediction values of .0226 and > .9999, respectively.

## Discussion

Our research goal was to identify which among the BDI-II, BHS, and Psychache scale was the optimal single screening measure for indicating suicide risk. In general, all three scales were significantly accurate in distinguishing between individuals in suicide at-risk groups, based on past and present suicidal behavior, from other individuals. However, the PAS displayed superior performance in identifying single time suicide attempters and cases with preexisting elevated risk. Specifically, in evaluating accuracy with AUCs, BHS values ranged from .65 (low) to .79 (moderate), BDI-II values varied from .67 (low) to .88 (moderate), and PAS values were .76 (moderate) to .94 (high). For each index of risk, the PAS achieved the highest observed AUC. As such, based on AUCs, in choosing among the three scales, the PAS is the preferred option.

AUCs and cut-scores based on the combination of sensitivity and specificity are one method of summarizing diagnostic efficiency. In particular, high sensitivity can be useful for *ruling out* a risk if an individual tests negative, and high specificity can have merit in *ruling in* a risk if a person tests positive (Akobeng, 2007a). There are, however, circumstances where other indices such as positive predictive power, negative predictive power, and relative risk ratios may be preferred. When those are used (Tables 3–4), however, the merit of the PAS relative to the BDI-II and BHS endures. Nevertheless, the small values for positive predictive power reemphasize the difficulty in identifying at-risk individuals. Further, depending on context, the costs of different screening errors (i.e., false positives, misses) can also be weighted and included with prevalence rates into utility analyses that consider more than just sensitivity and specificity.

Our optimal cut-score for identifying preexisting elevated risk with the best screening tool (PAS score ≥ 27) had a positive prediction value of .0226 meaning that 97.74% of screen-identified elevated risk cases were not actually at elevated risk. Nevertheless, this 2.26% rate is a substantial improvement over the base rate of 0.47%. In predicting ultimate death by suicide for psychiatric outpatients, positive prediction values of .0175 and .0138 have been indicated for the BDI and BHS, respectively (Beck et al., 1990) It is noteworthy that, unlike the BDI-II, the PAS contains no suicide-relevant content in its items and, yet, the PAS significantly surpasses the BDI-II in the accurate identification of both single time suicide attempters and cases with elevated risk. This finding aligns with Shneidman's (1993) assertion on the relevance of psychological pain, per se, as the cause of suicide.

Limitations to this research are present. Our sample was predominantly Caucasian, first-year university women. Replication with other ethnicities, sex-balanced samples, other age groups, and clinical samples can establish the generalizability of our results. Second, we focused on self-report measures for screening suicide risk. Comparisons with other methods such as clinician assessments, cognitive scales, and so on would be informative. Third, the study was cross-sectional, as such, we cannot examine the longitudinal outcomes of screening for suicide risk. Finally, although the added value of using all three measures could have been tested (e.g., through logistic regression), the incremental merit for each of these measures has been previously shown (Troister & Holden, 2013). Instead, given practical demands for a brief 5–7 min, valid assessment, we assessed which of the three measures had the most merit.

In summary, the BDI-II, BHS, and PAS were each shown to have value as brief, self-report screens for identifying preexisting levels of risk for suicide in students. Comparisons among scales highlighted the particular merit of the PAS. Further studies replicating these results with other populations will establish the generalizability of the PAS as a screen for suicide risk.

## References

Akobeng, A. K. (2007a). Understanding diagnostic tests 1: Sensitivity, specificity and predictive values. *Acta Paediatrica, 96,* 338–341. http://dx.doi.org/10.1111/j.1651-2227.2006.00180.x

Akobeng, A. K. (2007b). Understanding diagnostic tests 3: Receiver operating characteristic curves. *Acta Paediatrica, 96,* 644–647. http://dx.doi.org/10.1111/j.1651-2227.2006.00178.x

American College Health Association. (2000). *National college health assessment reference group report*. Baltimore, MD: American College Health Association.

Beck, A. T., Brown, G., Berchick, R. J., Stewart, B. L., & Steer, R. A. (1990). Relationship between hopelessness and ultimate suicide: A replication with psychiatric outpatients. *The American Journal of Psychiatry, 147,* 190–195. http://dx.doi.org/10.1176/ajp.147.2.190

Beck, A. T., & Steer, R. A. (1993). *Beck Scale for suicide ideation manual*. San Antonio, TX: Psychological Corporation.

Beck, A. T., Steer, R. A., Ball, R., & Ranieri, W. F. (1996). Comparison of Beck Depression Inventories –IA and –II in psychiatric outpatients. *Journal of Personality Assessment, 67,* 588–597.

Beck, A. T., Steer, R. A., & Brown, G. K. (1996). *Manual for the Beck Depression Inventory-II*. San Antonio, TX: Psychological Corporation.

Beck, A. T., Steer, R. A., & Carbin, M. G. (1988). Psychometric properties of the Beck Depression Inventory: Twenty-five years of evaluation. *Clinical Psychology Review, 8,* 77–100. http://dx.doi.org/10.1016/0272-7358(88)90050-5

Beck, A. T., Weissman, A., Lester, D., & Trexler, L. (1974). The measurement of pessimism: The hopelessness scale. *Journal of Consulting* *and Clinical Psychology, 42,* 861–865. http://dx.doi.org/10.1037/h0037562

Brown, G. K., Beck, A. T., Steer, R. A., & Grisham, J. R. (2000). Risk factors for suicide in psychiatric outpatients: A 20-year prospective study. *Journal of Consulting and Clinical Psychology, 68,* 371–377. http://dx.doi.org/10.1037/0022-006X.68.3.371

DeLisle, M. M., & Holden, R. R. (2009). Differentiating between depression, hopelessness, and psychache in university undergraduates. *Measurement & Evaluation in Counseling & Development, 42,* 46–63. http://dx.doi.org/10.1177/0748175609333562

Drum, D. J., Brownson, C., Burton Denmark, A., & Smith, S. E. (2009). New data on the nature of suicidal crises in college students: Shifting the paradigm. *Professional Psychology: Research and Practice, 40,* 213–222. http://dx.doi.org/10.1037/a0014465

Durham, T. W. (1982). Norms, reliability, and item analysis of the Hopelessness Scale in general psychiatric, forensic psychiatric, and college populations. *Journal of Clinical Psychology, 38,* 597–600. http://dx.doi.org/10.1002/1097-4679(198207)38:3<597::AID-JCLP2270380321>3.0.CO;2-6

Flett, G. L., Hewitt, P. L., & Heisel, M. J. (2014). The destructiveness of perfectionism revisited: Implications for the assessment of suicide risk and prevention of suicide. *Review of General Psychology, 18,* 156–172. http://dx.doi.org/10.1037/gpr0000011

Friedman, R. A. (2006). Uncovering an epidemic—Screening for mental illness in teens. *The New England Journal of Medicine, 355,* 2717–2719. http://dx.doi.org/10.1056/NEJMp068262

Hanley, J. A., & McNeil, B. J. (1983). A method of comparing the areas under receiver operating characteristic curves derived from the same cases. *Radiology, 148,* 839–843. http://dx.doi.org/10.1148/radiology.148.3.6878708

Holden, R. R., & Mehta, K. (1998). *The Psychache Scale*. Ontario, Canada: Authors.

Holden, R. R., Mehta, K., Cunningham, E. J., & McLeod, L. D. (2001). Development and preliminary validation of a scale of psychache. *Canadian Journal of Behavioural Science, 33,* 224–232. http://dx.doi.org/10.1037/h0087144

Joiner, T. E., Jr., Conwell, Y., Fitzpatrick, K. K., Witte, T. K., Schmidt, N. B., Berlim, M. T., . . . Rudd, M. D. (2005). Four studies on how past and current suicidality relate even when "everything but the kitchen sink" is covaried. *Journal of Abnormal Psychology, 114,* 291–303. http://dx.doi.org/10.1037/0021-843X.114.2.291

Mills, J. F., Green, K., & Reddon, J. R. (2005). An evaluation of the Psychache Scale on an offender population. *Suicide & Life-Threatening Behavior, 35,* 570–580. http://dx.doi.org/10.1521/suli.2005.35.5.570

Moœcicki, E. K. (1997). Identification of suicide risk factors using epidemiologic studies. *Psychiatric Clinics of North America, 20,* 499–517. http://dx.doi.org/10.1016/S0193-953X(05)70327-0

National Center for Health Statistics. (2014). *Health, United States, 2013: With special feature on prescription drugs*. Hyattsville, MD: U. S. Department of Health and Human Services.

National Institute of Mental Health. (2009). *Suicide in the U. S.: Statistics and prevention*. Retrieved from http://www.lb7.uscourts.gov/documents/08-42261.pdf

Paris, J. (2006). Predicting and preventing suicide: Do we know enough to do either? *Harvard Review of Psychiatry, 14,* 233–240. http://dx.doi.org/10.1080/10673220600968662

Robins, E., Murphy, G. E., Wilkinson, R. H., Jr., Gassner, S., & Kayes, J. (1959). Some clinical considerations in the prevention of suicide based on a study of 134 successful suicides. *American Journal of Public Health, 49,* 888–899. http://dx.doi.org/10.2105/AJPH.49.7.888

Roos, L., Sareen, J., & Bolton, J. M. (2013). Suicide risk assessment tools, predictive validity findings and utility today: Time for a revamp? *Neuropsychiatry, 3,* 483–495. http://dx.doi.org/10.2217/npy.13.60

Rosenberg, M. L., Davidson, L. E., Smith, J. C., Berman, A. L., Buzbee, H., Gantner, G., . . . Jobes, D. (1988). Operational criteria for the determination of suicide. *Journal of Forensic Sciences, 33,* 1445–1456.

Roy, A., Segal, N. L., Centerwall, B. S., & Robinette, C. D. (1991). Suicide in twins. *Archives of General Psychiatry, 48,* 29–32. http://dx.doi.org/10.1001/archpsyc.1991.01810250031003

Rudd, M. D., Joiner, T., & Rajad, M. H. (1996). Relationships among suicide ideators, attempters, and multiple attempters in a young-adult sample. *Journal of Abnormal Psychology, 105,* 541–550. http://dx.doi.org/10.1037/0021-843X.105.4.541

Shneidman, E. S. (1993). Suicide as psychache. *Journal of Nervous and Mental Disease, 181,* 145–147. http://dx.doi.org/10.1097/00005053-199303000-00001

Streiner, D. L., & Cairney, J. (2007). What's under the ROC? An introduction to receiver operating characteristics curves. *The Canadian Journal of Psychiatry, 52,* 121–128.

Thomson, W. (2012). Long-term follow-up of suicide in a clinically depressed community sample. *Journal of Affective Disorders, 139,* 52–55. http://dx.doi.org/10.1016/j.jad.2012.02.012

Troister, T., & Holden, R. R. (2010). Comparing psychache, depression, and hopelessness in their associations with suicidality: A test of Shneidman's theory of suicide. *Personality and Individual Differences, 49,* 689–693. http://dx.doi.org/10.1016/j.paid.2010.06.006

Troister, T., & Holden, R. R. (2012). A two-year prospective study of psychache and its relationship to suicidality among high-risk undergraduates. *Journal of Clinical Psychology, 68,* 1019–1027. http://dx.doi.org/10.1002/jclp.21869

Troister, T., & Holden, R. R. (2013). Factorial differentiation among depression, hopelessness, and psychache in statistically predicting suicidality. *Measurement & Evaluation in Counseling & Development, 46,* 50–63. http://dx.doi.org/10.1177/0748175612451744

(*Appendix follows*)

## Appendix

## The Psychache Scale[1]

| The following statements refer to your psychological pain, not your physical pain. By circling the appropriate number, please indicate how frequently each of the following occurs. | 1 = *never* | 2 = *sometimes* | 3 = *often* | 4 = *very often* | 5 = *always* |
|---|---|---|---|---|---|
| 1. I feel psychological pain. | 1 | 2 | 3 | 4 | 5 |
| 2. I seem to ache inside. | 1 | 2 | 3 | 4 | 5 |
| 3. My psychological pain seems worse than any physical pain. | 1 | 2 | 3 | 4 | 5 |
| 4. My pain makes me want to scream. | 1 | 2 | 3 | 4 | 5 |
| 5. My pain makes my life seem dark. | 1 | 2 | 3 | 4 | 5 |
| 6. I can't understand why I suffer. | 1 | 2 | 3 | 4 | 5 |
| 7. Psychologically, I feel terrible. | 1 | 2 | 3 | 4 | 5 |
| 8. I hurt because I feel empty. | 1 | 2 | 3 | 4 | 5 |
| 9. My soul aches. | 1 | 2 | 3 | 4 | 5 |
| Please continue this inventory using the following scale: | 1 = *strongly disagree* | 2 = *disagree* | 3 = *unsure* | 4 = *agree* | 5 = *strongly agree* |
| 10. I can't take my pain any more. | 1 | 2 | 3 | 4 | 5 |
| 11. Because of my pain, my situation is impossible. | 1 | 2 | 3 | 4 | 5 |
| 12. My pain is making me fall apart. | 1 | 2 | 3 | 4 | 5 |
| 13. My psychological pain affects everything I do. | 1 | 2 | 3 | 4 | 5 |

[1] From *The Psychache Scale*, by R. R. Holden and K. Mehta, 1998, Ontario, Canada: Authors. Reprinted with permission (Holden & Mehta, 1998).

Received June 21, 2014
Revision received February 25, 2015
Accepted February 26, 2015 ■